

James Shanley Pratt

Manchester

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Aspiring master's student of Chemical Engineering at Russell group Newcastle University currently achieving a 2.1 after third year, experienced in group and lab work having completed an plant design project on the creation of a methanol synthesis plant to a first standard.

Education:

University of Newcastle, MEng Chemical Engineering UG

Completed third year of university averaging a 2.1 which has developed my understanding in the core principals of chemical engineering with modules of Mathematics, Chemistry, Thermodynamics and Energy Transfers.

Multiple assessed coursework's and exams advanced my organisational skills and independent learning, improving the ability to meet time constraints under pressure.

Group work in the main modules of the course ranging from sugar beet refinery process (involving mass and energy balances across a production line) to jet engine analysis have given me experience of working constructively and effectively in a team, giving presentations (on Microsoft PowerPoint) and further improving communication skills between peers.

The plant design project in third year was to design a methanol synthesis plant, using natural gas as a raw material, reacting it to form synthesis gas and then to react it to form crude methanol. The methanol is then distilled to produce high quality methanol that is able to be sold on to manufacturers, as methanol is a raw material used in various processes. My individual part of the process was to fully design the topping, separating the light gases from the crude methanol in a distillation column.

2010-2017 St Bede's College Manchester

- A levels Chemistry (A), Mathematics (B), Physics(B)
- AS level Economics (B)
- GCSEs 10 A*- B
- EPQ (C)

The title of my voluntary EPQ was The Future Applications of Graphene, looking in more depth at water filtration, super capacitors, and sensors such as smart clothing (the first being the Little Black Dress with LEDs that respond to the wearer) or highly sensitive carbon monoxide sensors. Also, evaluating the advantages and disadvantages of Graphene over other products, for example its superconducting abilities at non-sub cooled temperatures and its troublesome production methods which include ensuring the quality of the product (most commonly through chemical

vapour deposition CVD) without making the product inaccessibly expensive to the market.

Work Experience:

Tennis Leader – West Heaton Tennis Club, Heaton Moor, coaching junior players during tennis sessions. I was also involved in recording and editing promotional videos to encourage disabled people to be more active involved in sport

2012-2014 Football Referee Manchester FA – This role boosted my confidence and ability to communicate clearly and effectively when under pressure in difficult situations.

ICT Skills:

MATLAB / Simulink – Used to create and analysis process such as multiple reactors, to see how various changes in feeds and reactor conditions change the outputs of the system. Using matlab to iteratively optimise reaction conditions of reactors in order to conditions to produce favourable outputs of the system.

Also, to create controller functions such as PID and internal model control systems to control conditions for desired setpoint tracking, disturbance rejection and noise control, implementing these and determining their effectiveness on the system.

Hysys – Using complex process software in order to simulate a process in order to determine the required feeds to achieve a desired output. This was used extensively for the third - year design project on the creation of methanol from synthesis gas.

CADassist Training Course - Completed Building and Information Modelling (BIM) course using Microsoft Revit (a 3D building and analysis software) enabling me to start project setups of buildings and analysis performance of heating, ventilation and air conditioning (HVAC) within it.

Microsoft Excel / PowerPoint / Word – All used extensively during university assessments.